

Classical Memory and Magic Are Interconvertible Resources in Fault-Tolerant Compilation

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Abstract

A fault-tolerant processor pays for a non-Clifford phase with magic states, and its classical controller holds, between rounds, a bounded number of bits. When a phase reaches the controller in several additive pieces, each piece must either be remembered until the last one arrives or be executed on arrival. We show that in the ancilla-free coordinatewise Clifford+ T synthesis model these two options are related by a fixed exchange rate α (committed T gates per bit of memory forgone) that does not depend on the problem size: rigorously $2 \leq \alpha \leq 3$. The lower bound $\alpha \geq 2$ follows from a new equidistribution estimate for Clifford+ T words near rotation cosets, whose only external input is the Ramanujan bound of Lubotzky–Phillips–Sarnak and Parzanchevski–Sarnak; the upper bound is attained by a simple fractional passthrough scheme with Ross–Selinger synthesis. The gap between 2 and 3 is exactly the square-root barrier of spectral equidistribution. Under an explicit conjecture (verified by exhaustive enumeration to T -count 22, with constants that agree with the Haar prediction) the rate is $\alpha = 3$ and commitment is quantized: touching a coordinate at all costs $2 \log_2(1/\varepsilon)$ T gates before it carries a single bit. Side information held downstream enters through a conditional entropy, which turns a remark of our earlier work into a theorem. Each bit of memory not held therefore costs between two and three magic states, and the constant 3 is the same constant as in the $3 \log_2(1/\varepsilon)$ cost of a single rotation.

1 Introduction

Two physical resources meet at the boundary between a fault-tolerant quantum processor and its classical control electronics. On the quantum side, every non-Clifford rotation consumes magic states, and the number of T gates is the dominant cost of a logical computation [1, 2]. On the classical side, the controller that resolves corrections, tracks the Pauli frame and issues instructions holds a bounded number of bits between rounds; frame tracking exists precisely because unbounded buffering is unavail-

able at that boundary [3, 4]. These are different resources with different exchange rates against everything else, and there is no a priori reason for them to be convertible into one another.

They become convertible when a phase arrives in pieces. A Trotter step, a randomized-compiling instance, a layerwise parametrized ansatz, or a phase-polynomial emitted by a scheduler presents the angle of a given rotation as a sum of shares delivered in different rounds. Whoever executes the stream then faces a choice at every share: hold it in classical memory until the sum is known and synthesize one rotation, or synthesize the share on arrival and consume the corresponding magic now. Our earlier work [5] established that the first option is charged in bits: any streaming process that is correct to projective error ε on all r -round dispersed streams of m commuting phases must carry, across the last inter-round cut, $\Omega(m \log(1/\varepsilon))$ bits of restart state or of committed output. The second option was there only estimated, through an empirical synthesis cost of about $3 \log_2(1/\varepsilon)$ T gates per rotation. The purpose of this paper is to make the second option a theorem and to determine the exchange rate.

Our object is the *committed magic* T_{pre} : the number of T gates executed before the last share of the stream has arrived. A process that holds every share in memory has $T_{\text{pre}} = 0$; a memoryless process must commit each share as it comes. The main results, stated for the ancilla-free coordinatewise Clifford+ T model of Sec. 2, are as follows.

- **Theorem 1** (unconditional). $T_{\text{pre}} \geq (r - 1)(m \log_2 K - S) - O(rm \log \log(1/\varepsilon))$, where S is the worst-input expected cut budget of [5] and $K = \Theta(1/\varepsilon)$: one committed T gate per bit not held, per round.
- **Theorem 2** (equidistribution). The number of Clifford+ T words of T -count $\leq k$ within ε of any coset $G_1 R_z(\theta) G_2$ of a rotation subgroup is at most $C_1 2^k \varepsilon^2 + C_2(k + 1)2^{k/2}$.
- **Theorem 3** (rate two). Consequently $T_{\text{pre}} \geq 2(r - 1)(m \log_2 K - S) - O(rm \log \log(1/\varepsilon))$ up to $O(\delta)$ corrections: every bit forgone costs at least two com-

mitted T gates.

- **Theorem 4** (achievability). A fractional passthrough scheme attains $T_{\text{pre}} = (r-1)(m-q)\tau(\varepsilon)$ at $S = q \log_2 Q$ for every q , hence rate $\tau(\varepsilon)/\log_2 K \rightarrow 3$ with Ross–Selinger synthesis, at every point of the frontier.
- **Conjecture 1 and Theorem 5** (rate three, quantization). Under a stated equidistribution conjecture—whose exponent is the codimension of a rotation coset in $\text{SU}(2)$, and which we verify numerically with Haar-exact constants—the rate is exactly 3, and a coordinate that receives any committed T gate receives at least $2 \log_2(1/\varepsilon) - O(1)$ of them before carrying its first bit.
- **Theorems 6 and 7** (side information). With side information Y available downstream, $m \log_2 K$ is replaced by the conditional entropy $H(X \mid \text{other rounds}, Y)$, with a one-shot version in terms of smooth entropies.

The two constants 2 and 3 have a common origin. A single-qubit Clifford+ T word of T -count t is one of $36 \cdot 2^t$ unitaries (Matsumoto–Amano), so a T gate carries at most one bit; a rotation coset is a curve in the three-dimensional group, so a word must first spend $\approx 2 \log_2(1/\varepsilon)$ gates to reach its ε -neighbourhood and then one gate per bit of angle. The Ross–Selinger cost $3 \log_2(1/\varepsilon)$ of a single rotation is the same count seen from the synthesis side. The rigorous rate 2 is what survives once the count near the coset is only controlled up to the square-root error of spectral methods (Remark 2); the conjectured rate 3 is what the exhaustive enumeration shows. We regard the statement “ $2 \leq \alpha \leq 3$, and the gap is the square-root barrier” as the honest form of the exchange law.

2 Setting

Family. We use the cyclic dispersed family of [5, Def. 13, 20; Cor. 7]. Fix a modulus $Q \geq 3$, step $\Delta = 2\pi/Q$, alphabet size $K = Q - 1$, and m $\mathbb{F}_2$ -independent Z -characters $P_1, \dots, P_m$ on $n \geq m$ qubits; let V be a fixed public Clifford with $VZ_jV^\dagger = P_j$. The aggregate is $x \in [K]^m$, $[K] = \{0, \dots, K-1\}$, and the target is

$$U_x = V \left(\bigotimes_{j \leq m} R_z(\Delta x_j) \otimes I \right) V^\dagger, \quad R_z(\theta) = e^{-i\theta Z/2}. \quad (1)$$

An r -round dispersed stream is a round-major sequence of shares $a^{(1)}, \dots, a^{(r)} \in \mathbb{Z}_Q^m$ with $\sum_t a^{(t)} \equiv x \pmod{Q}$. The *canonical sharing* $\mathcal{D}$ takes $a^{(1)}, \dots, a^{(r-1)}$ i.i.d. uniform on $\mathbb{Z}_Q^m$ and $a^{(r)} = x - \sum_{t < r} a^{(t)}$. The accuracy satisfies $0 < \varepsilon < \sin(\pi/2Q)$, so that distinct grid rotations are separated by $2 \sin(\pi/2Q) > 2\varepsilon$ in the projec-

tive distance $d_{\text{proj}}(A, B) = \inf_\phi \|A - e^{i\phi} B\|$; the tuned modulus $Q_\varepsilon = \lfloor \pi/(2 \arcsin 2\varepsilon) \rfloor$ of [5, Cor. 7] gives $\log_2 K_\varepsilon = L - 0.35 + o(1)$, where throughout

$$L := \log_2(1/\varepsilon). \quad (2)$$

Output model. The *coordinatewise ancilla-free Clifford+ T model* (CW) is the one realized by per-rotation synthesis pipelines [5, Rem. 6]: the write-once output is a list of records (j, g) with $j \in [m]$ and $g \in \{H, S, T\}$, the implemented unitary is $V(\bigotimes_j W_j)V^\dagger$ where W_j is the ordered product of the gates recorded on coordinate j , and the T -count is the number of T records. We say the output is *correct* if $d_{\text{proj}}(W_j, R_z(\Delta x_j)) \leq \varepsilon$ for every j . By Lemma 1, correctness of the whole unitary in the sense of [5] implies coordinatewise correctness, so every lower bound below also holds under the total criterion.

Process and committed magic. The process is a compiler in the charged one-pass model of [5, Def. 9, 21]: it reads the stream once, may use arbitrary RAM and a random tape whose state is part of every restart snapshot, and $S := \max_{\text{input}} \mathbb{E}_\rho \max_c B_{\text{cut}}(c)$ is the worst-input expected snapshot size. It is required to be correct with probability $\geq 1 - \delta$ on every x and every valid stream, $\delta < \frac{1}{2}$. Let T_t be the number of T records emitted while reading round t , $T_{\text{pre}} := \sum_{t < r} T_t$ the *committed magic*, and $T_{\text{out}} = T_{\text{pre}} + T_r$. Bars denote expectation over the tape, the canonical sharing and a uniform aggregate. We write $L_\delta(m, K) := (1 - \delta)m \log_2 K - h_2(\delta)$, $\ell_\delta := (1 - \delta) \log_2 K - h_2(\delta)$, and $\rho_m(\bar{T}) := m \log_2(e(1 + \bar{T}/m)) + 2 \log_2(1 + \bar{T}) + 1$.

3 Three lemmas

Lemma 1 (Localization). *For unitaries $A = \bigotimes_j A_j$, $B = \bigotimes_j B_j$ on the same tensor factors, $\max_j d_{\text{proj}}(A_j, B_j) \leq d_{\text{proj}}(A, B) \leq \sum_j d_{\text{proj}}(A_j, B_j)$. For 2×2 unitaries, $d_{\text{proj}}(A, B) = \sqrt{2 - |\text{tr}(B^\dagger A)|}$.*

Lemma 2 (Matsumoto–Amano count). *The number of single-qubit Clifford+ T unitaries modulo phase of minimal T -count exactly t is $n_1(0) = 24$ and $n_1(t) = 36 \cdot 2^t$ for $t \geq 1$; hence $\sum_{s \leq t} n_1(s) = 72 \cdot 2^t - 48$, and the number of m -tuples with total minimal T -count at most T is $N_m^{\leq}(T) \leq 36^m 2^T \binom{T+m}{m}$.*

Lemma 3 (Entropy versus T -count). *Let $M = (\bar{W}_1, \dots, \bar{W}_m)$ be a random m -tuple of single-qubit Clifford+ T unitaries and $T \geq \sum_j t_{\min}(\bar{W}_j)$ a random integer. Then $H(M) \leq \bar{T} + m \log_2 36 + \rho_m(\bar{T})$.*

Proofs are in Appendix A. Lemma 2 is the source of the exchange rate: a T gate carries at most $\log_2 2 = 1$ bit.

4 Committed magic: the unconditional bound

Theorem 1 (Committed magic). *For every process as in Sec. 2 and every round $t \in [r-1]$,*

$$S + \bar{T}_t + m \log_2 36 + \rho_m(\bar{T}_t) \geq L_\delta(m, K). \quad (3)$$

Consequently

$$T_{\text{pre}}^- \geq (r-1)[L_\delta(m, K) - S - m \log_2 36] - (r-1)\rho_m\left(\frac{T_{\text{pre}}^-}{r-1}\right), \quad (4)$$

and for the tuned family and $\delta = 0$, $T_{\text{pre}} \geq (r-1)(m \log_2 K_\varepsilon - S) - O(rm \log \log(1/\varepsilon))$.

Proof. Fix t . Let $A := a^{(t)}$, $\text{Past} := (a^{(s)})_{s < t}$, $\text{Fut} := (a^{(s)})_{s > t}$ (which includes round r), and $Z := (\text{Past}, \text{Fut}, \rho)$.

(i) $H(A | Z) = m \log_2 K$. Under $\mathcal{D}$ the joint law of $(a^{(1..r-1)}, x)$ is $Q^{-m(r-1)} P_X(x)$ with $a^{(r)}$ determined. Conditioning on $(\text{Past}, \text{Fut})$ fixes $c := \sum_{s \neq t} a^{(s)}$ and $x = A + c$, so the conditional law of A is proportional to $P_X(A + c)$, i.e. uniform on the translate $[K]^m - c$. The tape is independent of the stream.

(ii) *Message and Fano.* Let $M_t := (\bar{W}_1^{(t)}, \dots, \bar{W}_m^{(t)})$ be the tuple of unitaries (modulo phase) implemented by the gate segments emitted on the respective coordinates during round t , and σ_t the restart snapshot at the cut after round t . Given ρ the process is deterministic; the segments emitted before round t are a function of (Past, ρ) , and those emitted after the cut are a function of (σ_t, Fut) by the restart semantics of [5, Def. 9]. Hence the final unitary on coordinate j , $U_j^{<t} \bar{W}_j^{(t)} U_j^{>t}$, is a function of (M_t, σ_t, Z) . On success it is within ε of $R_z(\Delta x_j)$, and nearest-grid decoding recovers x_j because neighbouring grid rotations are more than 2ε apart; thus $A = x - c$ is recovered with error probability $\leq \delta$. Fano's inequality gives $H(A | M_t, \sigma_t, Z) \leq h_2(\delta) + \delta \log_2(K^m - 1)$, i.e. $I(A; M_t, \sigma_t | Z) \geq L_\delta(m, K)$.

(iii) *Message size.* $I(A; M_t, \sigma_t | Z) \leq H(\sigma_t) + H(M_t)$. The snapshot serialization is prefix-free [5, Lem. 2], so $H(\sigma_t) \leq \mathbb{E}|\sigma_t| \leq S$, the worst-input expectation dominating the average over inputs. The number T_t of T records emitted in round t is at least $\sum_j t_{\min}(\bar{W}_j^{(t)})$, so Lemma 3 gives $H(M_t) \leq \bar{T}_t + m \log_2 36 + \rho_m(\bar{T}_t)$. This proves (3); summing over t and using concavity of ρ_m proves the rest. $\square$

Remark 1 (Multiple passes). For p forward passes, the alternating-simulation transcript of [5, Def. 21] (all $2p-1$ crossing snapshots together with the unitary tuple committed while scanning rounds $< r$ in any pass) determines the output, and the same argument gives $\Sigma_p + \bar{T}_A + m \log_2 36 + \rho_m(\bar{T}_A) \geq L_\delta$ with $\Sigma_p \leq (2p-1)S$. The

per-round splitting, and with it the factor $r-1$, is a one-pass phenomenon: a process that rereads round t after seeing round r need not commit anything during round t .

Theorem 1 is the bit-level theorem of [5] with Lemma 2 in place of Kraft's inequality; its content is that the counting rate $\log_2 2 = 1$ is the worst case. The rest of the paper is about the fact that the true rate is larger.

5 Words near rotation cosets

Arithmetic structure. Single-qubit Clifford+ T modulo phase is the full $\{(\sqrt{2})\}$ -arithmetic group Γ of the maximal order

$$\mathcal{O} = \mathcal{O}_K \oplus \mathcal{O}_K \frac{1+i}{\sqrt{2}} \oplus \mathcal{O}_K \frac{1+j}{\sqrt{2}} \oplus \mathcal{O}_K \frac{1+i+j+k}{2}, \quad \mathcal{O}_K = \mathbb{Z}[\sqrt{2}], \quad (5)$$

of the definite quaternion algebra $D = (-1, -1)_{\mathbb{Q}(\sqrt{2})}$ [8, 9]. D splits at the prime $P = (\sqrt{2})$; the completion $K_P = \mathbb{Q}_2(\sqrt{2})$ has residue field $\mathbb{F}_2$, and Γ acts on the 3-regular Bruhat–Tits tree X_P with vertex stabilizer $U = \mathcal{O}^\times / \mathcal{O}_K^\times \cong C_{24}$, the Clifford group [9, §4.1.3]. *T-count equals tree distance:* the sphere of radius t has $3 \cdot 2^{t-1}$ vertices, and $24 \cdot 3 \cdot 2^{t-1} = 36 \cdot 2^t$ is exactly Lemma 2. Let $\Lambda_k \subset \Gamma$ be the ball of T -count $\leq k$, $N_k = |\Lambda_k| = 72 \cdot 2^k - 48$, and for a left- U -invariant $h \in L^2(\text{PU}(2))$ let $T_j h(g) := \sum_{\gamma U: d(\gamma v_0, v_0) = j} h(\gamma^{-1} g)$ be the Hecke operator of level j .

Input (R). On $L_0^2(U \setminus \text{PU}(2))$ one has $\|T_j\| \leq (j+1)2^{j/2}$ [9, Eq. (3.10)]. This is the Ramanujan bound: by Jacquet–Langlands the relevant automorphic representations of $D^\times$ are cuspidal Hilbert modular forms over $\mathbb{Q}(\sqrt{2})$, which are tempered [11, 12, 13]. The non-backtracking version [9, Eqs. (5.2)–(5.3)] gives the same bound directly for the Matsumoto–Amano syllables HT and SHT .

Theorem 2 (Tube bound). *There are absolute constants $C_1 \leq 288$ and C_2 such that for all $G_1, G_2 \in \text{SU}(2)$, all $\varepsilon \in (0, 1/8]$ and all $k \geq 0$, with $T_\varepsilon(C) := \{W : \exists \theta, d_{\text{proj}}(W, G_1 R_z(\theta) G_2) \leq \varepsilon\}$,*

$$\#(\Lambda_k \cap T_\varepsilon(C)) \leq C_1 2^k \varepsilon^2 + C_2 (k+1) 2^{k/2}. \quad (6)$$

The proof (Appendix B) reduces the tube to a spherical cap via the Hopf fibration whose fibre is the coset, and bounds the count by the Hecke operator through a rank-one sup-norm estimate.

Remark 2 (The square-root barrier). Any majorant of a cap of radius ε on S^2 has spectral mass up to degree $\gtrsim 1/\varepsilon$, so the error term $(k+1)2^{k/2}$ in (6) cannot be reduced by this method. It exceeds the main term $2^k \varepsilon^2$ exactly when $k < 4L - 2 \log_2(k+1)$, i.e. throughout the range $k \in [2L, 3L]$ in which synthesized rotations live. The true count in this range (Sec. 9) is the main term; proving that is the content of Conjecture 1.

Corollary 1 (Alphabet profile). *In the setting of Theorem 1, conditional on $(\sigma_t = s, Z = z)$, the unitary $\bar{W}_j^{(t)}$ committed on coordinate j during round t lies in $T_\varepsilon(C_{s,z,j})$ for the coset with $G_1 = Y_j(s, \text{Fut})^\dagger$ (the after-cut unitary) and $G_2 = (U_j^{<t})^\dagger$; the grid shift $R_z(\Delta c_j)$ is absorbed into θ . Hence, with $n(\tau) := C_1 2^\tau \varepsilon^2 + C_2(\tau + 1)2^{\tau/2}$, $\#\{\bar{W}_j^{(t)} : t_{\min} \leq \tau\} \leq n(\tau)$ and $\log_2 n(\tau) \leq \tau/2 + \log_2(\tau + 1) + C$ for $\tau \leq 4L - 2\log_2(\tau + 1) - C'$.*

Theorem 3 (Rate two). *Under (R), for every round t ,*

$$\bar{T}_t \geq 2[(1 - 2\delta)m \log_2 K - 2m h_2(\delta) - S] - O(m \log \log(1/\varepsilon)), \quad (7)$$

hence $T_{\text{pre}}^- \geq 2(r - 1)[(1 - 2\delta)m \log_2 K - 2m h_2(\delta) - S] - O(rm \log \log(1/\varepsilon))$.

The proof (Appendix C) shows that, given the snapshot and the other rounds, a committed word of T -count τ carries at most $\tau/2 + O(\log L)$ bits about its coordinate whenever $\tau \leq 4L$, and that words of larger cost cannot help because a coordinate carries at most $\log_2 K \leq 2L$ bits. Theorem 3 bounds a rate; it contains no statement about quantization, because the profile $(\tau + 1)2^{\tau/2}$ is not $O(1)$ at $\tau \leq 2L$.

6 The achievable frontier

Theorem 4 (Fractional passthrough). *For every $q \in \{0, \dots, m\}$ and every single-rotation synthesizer of cost $\tau(\varepsilon)$ per rotation to projective error ε , there is a deterministic zero-error one-pass CW process with*

$$\begin{aligned} S &\leq \lceil q \log_2 Q \rceil + O(\log(mQr)), \\ T_t &\leq (m - q) \tau(\varepsilon/r) \quad (t < r), \\ T_{\text{out}} &\leq (m + (r - 1)(m - q)) \tau(\varepsilon/r). \end{aligned} \quad (8)$$

Proof. Keep running sums modulo Q of q chosen coordinates (the block compiler of [5, Prop. 10] with $p = 1$); emit every share of the remaining coordinates as it arrives, synthesized to ε/r (a zero share costs nothing); in round r emit the aggregate rotation of each stored coordinate and the last share of each committed one. The coordinatewise error is at most ε by the right inequality of Lemma 1 and unitary invariance of d_{proj} . $\square$

Corollary 2. *Along the whole frontier, $T_{\text{pre}}^{\text{ach}}/[(r - 1)(m \log_2 K - S)] \leq \tau(\varepsilon/r)/\log_2 K \cdot (1 + o(1))$. With Ross-Selinger synthesis, $\tau = 3L + O(\log L)$ typical and $4L + O(1)$ worst [6], this is within a factor $\rightarrow 3$ of Theorem 1 and $\rightarrow 3/2$ of Theorem 3. Under Conjecture 1 it matches Theorem 5 to leading order in L .*

The endpoint gap $\log_2 Q$ versus $\log_2 K$ is inherited from [5, Prop. 10–11].

7 Rate three and the quantum of commitment

Conjecture 1 (H). *There are constants c_0, c such that for all $G_1, G_2 \in \text{SU}(2)$, all $\varepsilon \in (0, 1/8]$, all Q with $\varepsilon < \sin(\pi/2Q)$ and all $\tau \geq 0$,*

$$\begin{aligned} \#\{W : t_{\min}(W) \leq \tau, \exists d \in \mathbb{Z}_Q : \\ d_{\text{proj}}(W, G_1 R_z(\frac{2\pi d}{Q}) G_2) \leq \varepsilon\} &\leq c_0 + c 2^\tau \varepsilon^\beta, \quad \beta = 2. \end{aligned} \quad (9)$$

The exponent is the codimension of a rotation coset in the three-dimensional group. Haar measure predicts $c = 12$ for the tuned grid and $c_0 \leq 8$ (the powers of T in the (G_1, G_2) frame); Sec. 9 confirms both. By Remark 2 the conjecture lies beyond (R): in the parametrization of Appendix A it asserts that among the $\xi = tt^\dagger \in \mathbb{Z}[\sqrt{2}]$ in a thin strip, those with $2^k - \xi$ also a relative norm have density $2^{-k+o(k)}$ —a shifted-norm sieve at the ramified prime $(1 - \omega)$.

Theorem 5 (Rate three; quantization). *Assume Conjecture 1. Let $B_t := m\ell_\delta - S$ and $\kappa := 1 + (\beta L - \log_2 c - 1)/\log_2 K$. For every round t ,*

$$\begin{aligned} \bar{T}_t &\geq \kappa B_t - \kappa(1 + \log_2(c_0 + 1))m - 2m \log_2(1 + \bar{T}_t/m) \\ &\quad - 2m(h_2(\delta) + \delta \log_2 K), \end{aligned} \quad (10)$$

$$\bar{A}_t \geq \frac{B_t - m(1 + \log_2(c_0 + 1)) - m(h_2(\delta) + \delta \log_2 K)}{\log_2 K}, \quad (11)$$

where $\bar{A}_t$ is the expected number of coordinates receiving a T gate in round t . With $\beta = 2$, $c = 12$ and the tuned family, $\kappa \rightarrow 3$: $T_{\text{pre}}^- \geq 3(r - 1)(m \log_2 K - S)(1 - o(1)) - O(rm \log \log(1/\varepsilon))$, and every coordinate that is touched costs at least $2L - \log_2 12 - 1$ T gates before it carries a single bit.

The proof is in Appendix C. The structural content is that memory cannot be shed a few bits at a time within a coordinate: the first $2L$ committed gates on a coordinate buy nothing, and the optimal way to trade memory for magic is to commit whole coordinates, which is what fractional passthrough does.

Lemma 4 (Identity coset). *If $2\varepsilon^2 4^k < 1$, i.e. $k < L - \frac{1}{2}$, the only Clifford+ T unitaries with denominator exponent $\leq k$ within ε of some $R_z(\theta)$ are the eight powers of T .*

Lemma 4 (Appendix A) is an elementary instance of the fee: at the identity coset the first L gates buy nothing, unconditionally. The enumeration shows the true fee there is $2L + O(1)$: the least T -count ≥ 2 at which some word comes within ε of a grid rotation is 10, 10, 12, 17, 18 for $L = 4, \dots, 8$, i.e. between $2L$ and $2L + 3$.

8 Side information

Theorem 6 (Side information). *Let $X \sim P_X$ on $[K]^m$, let the sharing have any joint law with X , and let side information Y be available to the process from the start. Then for every round t ,*

$$\begin{aligned} S + \bar{T}_t + m \log_2 36 + \rho_m(\bar{T}_t) \\ \geq H(X \mid \text{Past}, \text{Fut}, Y) - h_2(\delta) - \delta \log_2(K^m - 1). \end{aligned} \quad (12)$$

Proof. Theorem 1 with $Z := (\text{Past}, \text{Fut}, Y, \rho)$; the tape is independent of (stream, Y) , and $a^{(t)} \leftrightarrow X$ is a bijection given the other rounds. $\square$

Corollary 3. (a) *Uniform X , canonical sharing, trivial Y recovers Theorem 1.* (b) *If Y determines the earlier shares (the process holds the sharing or twirl seed), the right side is 0: no memory and no committed magic. This is the boundary of the design rule “randomize and lower late” of [5].* (c) *If the shares are generated from a λ -bit uniform seed unknown to the process by an injective map, then $H(X \mid V) \geq H(X) - H(V) + H(V \mid X) \geq \lambda - m \log_2(Q/K)$; this is the precise form of [5, Rem. 9].* (d) *For non-product sources Theorems 3 and 5 hold with $m\ell_\delta$ replaced by $\sum_j H(A_j \mid Z) - m(h_2(\delta) + \delta \log_2 K)$ and S by $S + \text{TC}(A \mid Z)$, since $\sum_j I(A_j; \sigma \mid Z) \leq I(A; \sigma \mid Z) + \text{TC}(A \mid Z)$.*

Theorem 7 (One-shot). *Let the process be deterministic with worst-case resources $S_{\max}$, $T_{\max}$, correct with probability $\geq 1 - \delta$ over $(X, \text{sharing}, Y)$, and $r = 2$. With the set-smoothed entropy $H_0^\delta(U \mid VY) := \min_{\Omega: \Pr[\Omega] \geq 1-\delta} \log_2 \max_{v,y} |\{u : (u, v, y) \in \Omega\}|$ of Renner–Wolf [14],*

$$\begin{aligned} S_{\max} + T_{\max} + m \log_2 72 + m \log_2(e(1 + T_{\max}/m)) \\ \geq H_0^\delta(U \mid V, Y) \geq H_{\max}^\delta(U \mid V, Y). \end{aligned} \quad (13)$$

For i.i.d. coordinates, $H_{\max}^\delta(U^m \mid V^m Y^m) = mH(U \mid VY) + \sqrt{mV} \Phi^{-1}(1-\delta) + O(\log m)$ [15], V the conditional information variance. The Shannon form (Theorem 6) applies to expected resources and the one-shot form to worst-case resources; the two are not to be mixed.

9 Numerical evidence

We enumerated all single-qubit Clifford+ T unitaries of T -count ≤ 22 (1.5×10^8 words) through the Matsumoto–Amano normal form, with the 24 right Cliffords verified by closure.

Conjecture 1 (Fig. 1). For each $t \leq 22$, $\varepsilon \in \{2^{-4}, \dots, 2^{-8}\}$ with tuned $Q_\varepsilon \in \{12, 25, 50, 100, 201\}$, and three coset pairs (identity and two Haar-random

Table 1: Grid counts on Haar coset pair 0; Haar prediction $12 \cdot 2^{t-2L}$ in parentheses.

t	$\varepsilon = 2^{-4}$	2^{-5}	2^{-6}	2^{-7}	2^{-8}
12	187 (192)	48 (48)	11 (12)	2 (3)	0 (0.75)
15	1470 (1536)	372 (384)	100 (96)	23 (24)	4 (6)
19	23426 (24576)	6104 (6144)	1487 (1536)	338 (384)	77 (96)
22	187707 (196608)	48772 (49152)	12321 (12288)	3039 (3072)	786 (768)

(G_1, G_2) , we counted words within ε of some grid rotation $G_1 R_z(2\pi d/Q) G_2$ and words within ε of the whole coset. A least-squares fit over the 136 cells with count ≥ 30 gives

$$\log_2 N = 1.001 t - 2.007 L + \text{const}, \quad (14)$$

i.e. exponents (1, 2) to three decimals. The constants are not fitted: at $t = 22$, pooling the two Haar pairs, the grid counts give $N/(2^t \varepsilon^2) = 11.47, 11.96, 11.99, 12.16, 12.14$ for $\varepsilon = 2^{-4}, \dots, 2^{-8}$ and the tube counts 35.98, 35.98, 36.02, 36.20, 36.43. Both are the exact Haar predictions, 12 and 36. Over the whole table, restricted to the 32 Haar-pair cells whose *expected* count exceeds 3000, the grid ratios lie in $[11.3, 12.4]$ and the tube ratios in $[35.1, 36.3]$; the cut must be on the expected and not the observed count, since selecting cells by observed count conditions on the upper Poisson tail. What the three extra T -counts buy is the small- ε end of the table, where the expected counts were previously of order one: at $t = 19$ the $\varepsilon = 2^{-8}$ cell held 77 words against a prediction of 96 (20% low, and Poisson-noisy), whereas at $t = 22$ it holds 786 against 768 (2% high). Under Haar measure the squared off-diagonal entry $|b|^2$ is uniform on $[0, 1]$, so the tube $\{|b|^2 \leq \varepsilon^2\}$ has measure ε^2 and $36 \cdot 2^t \varepsilon^2$ of the $36 \cdot 2^t$ words of T -count t are expected in it; the grid acceptance region $\{s + \varphi^2/4 \leq \varepsilon^2\}$ in the (transverse, angular) coordinates occupies one third of the tube at grid spacing 8ε , giving $12 \cdot 2^t \varepsilon^2$. The words are therefore Haar-equidistributed near rotation cosets to within statistical error wherever the expected count is at least of order one. At the identity coset the count is exactly 0 for $2 \leq t \leq 9$ at $L = 4$, with onset at $t = 10 = 2L + 2$: arithmetic cosets are *below* Haar at low cost, which is the fee.

Exact optimal synthesis and the frontier (Fig. 2).

The same enumeration yields, for $L \leq 6$, the minimal T -count $\tau_{\min}(d)$ of a word within ε of every grid rotation $R_z(2\pi d/Q)$; this is the exact optimum, independent of factoring. At $L = 5$ ($Q = 25$) the mean over $d \neq 0$ is 11.83 ($3L = 15$), at $L = 6$ ($Q = 50$) it is 15.59 ($3L = 18$). Fractional passthrough with canonical sharing ($m = 64$, $r = 2$) then has $S = q \log_2 Q$ and $T_{\text{pre}} = (m - q) \mathbb{E}_u \tau_{\min}(u)$, with achieved slope $\mathbb{E} \tau / \log_2 K = 2.48$ ($L = 5$) and 2.72 ($L = 6$), approaching 3 from below as $(3L - 2.5)/(L - 0.35)$. A partial-commitment variant

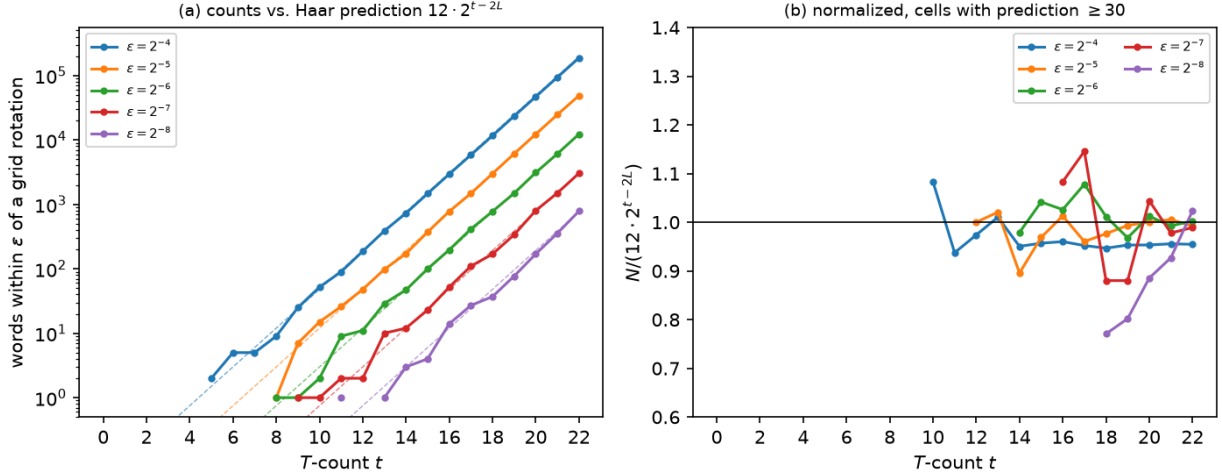


Figure 1: Number of Clifford+ T words of T -count exactly t within ϵ of a grid rotation on a Haar-random coset $G_1 R_z(\theta) G_2$. (a) Counts (markers) against the Haar prediction $12 \cdot 2^{t-2L}$ (dashed). (b) The same counts divided by that prediction, plotted where the prediction is at least 30; every ϵ has reached the Haar value by $t = 22$. Exhaustive enumeration to $t = 22$ (1.5×10^8 words).

that commits the low ℓ bits of a share as a rotation and stores the rest costs 5.0, 3.4, 3.0, 2.9 T gates per bit shed for $\ell = 1, \dots, 4$ at $L = 5$, and 8.5, 6.1, 4.6, 3.8, 3.0 for $\ell = 1, \dots, 5$ at $L = 6$: strictly worse than passthrough and worsening as ℓ decreases, which is the quantum of commitment seen at exact optimality. The additive terms $m \log_2 36$ and ρ_m of Theorems 1 and 3 are not small at $L \leq 6$; Fig. 2(a) tests slopes, not intercepts.

Chemistry scale (Fig. 2(b)). Exhaustive enumeration stops at $L \leq 6$, so at a chemically relevant accuracy we replace $\tau_{\min}$ by Ross–Selinger synthesis (`gridsynth`, whose ϵ is the up-to-phase operator norm and so is exactly d_{proj}). At $\epsilon = 10^{-10}$ the tuned modulus is $Q_\epsilon = 7853981633$ and $\log_2 K_\epsilon = 32.871$. Under canonical sharing the share u is uniform on $\mathbb{Z}_Q$ and the frontier depends on it only through $\mathbb{E}_u \tau(u)$, so we sample rather than synthesize all Q_ϵ rotations: 20 000 uniform u give $\mathbb{E}_u \tau = 102.32 \pm 0.02$ (s.d. 2.16, range 88–113) against $3L = 99.66$, hence an achieved slope $\mathbb{E} \tau / \log_2 K = 3.113 \pm 0.001$. Exceeding 3 slightly is expected — $\tau_{\text{RS}} = 3L + O(\log L)$, and Corollary 2 asserts only the limit. With $m = 10^4$ and $r = 2$ this is $T_{\text{pre}} = 1.023 \times 10^6$ committed T gates at $S = 0$; both axes are linear in m , so $m = 10^5$ is the same line scaled tenfold. Partial commitment of the low ℓ bits now costs 61, 46, 35, 28, 22, 19, 16, 14 T gates per bit shed for $\ell = 1, \dots, 8$: the quantum of commitment persists at scale, and is an order of magnitude above the passthrough slope, because a share whose high bits are known is still a generic angle of the fine grid.

10 Model dependence

Table 1 summarizes what changes with the synthesis model. Two entries deserve comment. In the phase-gradient model the exchange is bit-granular: adding the high bits of a share into the top bits of the gradient register is an exact operation whose cost is linear in the number of bits, so the fee of the CW model disappears; the catalytic gradient state is, in this sense, a device that has paid the fee once for all rotations. In the fully general n -qubit model the counting argument degenerates, because a T gate on n qubits can be any of $2 \cdot 4^n$ Pauli rotations; whether an $\Omega(m \log(1/\epsilon))$ committed-magic bound survives ancilla-assisted synthesis is open, and is equivalent to a data-structure lower bound for diagonal unitaries. Magic monotones do not help here: they bound the magic of the final unitary, which is $\Theta(m)$, not the magic committed before the information arrived.

11 Discussion

The law we have established is a conservation statement about two currencies that are usually accounted separately. In the model where rotations are synthesized coordinate by coordinate without ancillas, each bit of aggregate phase information that the controller does not retain costs at least two magic states before the phase is fully known, and three suffice. The rigorous lower bound 2 is what the Ramanujan bound for the octahedral golden gates yields once the sup-norm loss of pointwise counting is paid; the upper bound 3 is Ross–Selinger. That the two constants bracket the same number 3 as the cost of a single rotation is not a coincidence: both are volume

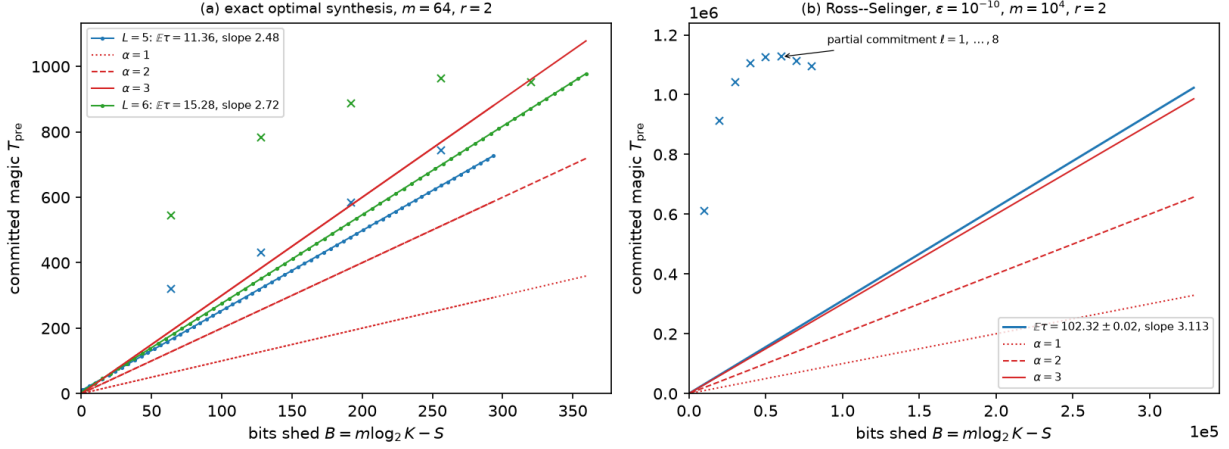


Figure 2: Achievable frontier of fractional passthrough ($r = 2$) against the leading-order lines of slope 1 (Theorem 1), 2 (Theorem 3) and 3 (Theorem 5). (a) Exact minimal T -counts from the enumeration, $m = 64$, $L = 5, 6$ — an optimum no synthesis algorithm can beat. (b) Ross-Selinger synthesis at the chemistry scale $\varepsilon = 10^{-10}$, $m = 10^4$, from 20 000 sampled grid rotations. Crosses: partial commitment of the low ℓ bits of each share.

counts in $SU(2)$, the one transverse and the other along the coset, and Conjecture 1 states that the words do not know the difference. The enumeration says they do not.

Three consequences are worth stating plainly. First, the classical memory of a fault-tolerant control stack has a magic-state price, and resource estimators can report the committed magic T_{pre} of a schedule as a first-class quantity, with $T_{\text{pre}} \geq 2(r-1)(m \log_2 K - S)$ as a floor. Second, memory cannot be traded for magic in small units within a rotation; the trade is quantized in whole rotations, and a scheduler that must shed memory should shed whole coordinates. Third, side information changes everything at the level of a conditional entropy: a process that holds the seed of the randomization pays nothing, and one that knows only that a λ -bit seed exists pays λ rather than $m \log_2 K$.

Parzanchevski and Sarnak show that some single-qubit unitaries need T -count $4L$ while all are reachable at $6L$ [9]; the committed-magic frontier is a statement about the typical rather than the worst case, and it sits between the rigorous and the conjectural covering exponents in the same way. Closing the gap between 2 and 3 is an arithmetic problem about lattice points near a curve below the square-root barrier, and we leave it open.

Acknowledgments

(To be added.)

A Proofs of the lemmas

Proof of Lemma 1. By [5, Lem. 1], $d_{\text{proj}}(A, B) = 2 \sin(\Theta/4)$ where $\Theta \in [0, 2\pi]$ is the length of the shortest closed arc containing $\text{spec}(B^\dagger A)$. The spectrum of

$\bigotimes_j B_j^\dagger A_j$ consists of the products of eigenvalues. Fix j and two eigenvalues λ, λ' of $B_j^\dagger A_j$ at circular distance $\theta_j \leq \pi$; multiplying both by a fixed product μ of eigenvalues of the other factors gives two points of the joint spectrum at circular distance θ_j , and any closed arc containing both has length at least θ_j . Hence $\Theta \geq \theta_j$, and since $\sin$ increases on $[0, \pi/2]$, $d_{\text{proj}}(A, B) \geq 2 \sin(\theta_j/4) = d_{\text{proj}}(A_j, B_j)$. For the right inequality replace one factor at a time: $\|\bigotimes A_j - e^{i \sum \phi_j} \bigotimes B_j\| \leq \sum_j \|A_j - e^{i \phi_j} B_j\|$, then take infima. For the closed form, with eigenphases a, b of $B^\dagger A$ one has $|\text{tr}|^2 = 2 + 2 \cos \Theta$ and $2 \sin(\Theta/4) = \sqrt{2 - 2 \cos(\Theta/2)} = \sqrt{2 - |\text{tr}|}$, using $\cos(\Theta/2) \geq 0$ for $\Theta \leq \pi$. $\square$

Proof of Lemma 2. Every Clifford+ T unitary has a unique normal form $(T|\epsilon)(HT|SHT)^*C$ with C one of the 24 Cliffords modulo phase [7], and the normal form is T -optimal [8]. Strings with exactly t letters T number 2^{t-1} (initial T present) plus 2^t (absent), times 24. For tuples, $\prod_j n_j(t_j) \leq 36^m 2^{\sum_j t_j}$ and the number of (t_j) with $\sum t_j \leq T$ is $\binom{T+m}{m}$. $\square$

Proof of Lemma 3. Encode M by the Elias- γ code of T (at most $2 \log_2(T+1)+1$ bits) followed by an index among $N_m^{\leq}(T)$ tuples; $H(M)$ is at most the expected length of this prefix code. Use $\log_2 \binom{T+m}{m} \leq m \log_2(e(1+T/m))$, which is concave in T , and Jensen. $\square$

Proof of Lemma 4. Write the unitary as $2^{-k/2} \begin{pmatrix} u & -t^\dagger \omega^l \\ t & u^\dagger \omega^l \end{pmatrix}$ with $u, t \in \mathbb{Z}[\omega]$, $\omega = e^{i\pi/4}$, $uu^\dagger + tt^\dagger = 2^k$ [8]. By the closed form of Lemma 1, closeness to a diagonal unitary forces $|t|^2 \leq 2\varepsilon^2 2^k$. The Galois conjugate $\sqrt{2} \mapsto -\sqrt{2}$ of the norm equation gives $|t^\bullet|^2 \leq 2^k$. Hence $\xi := tt^\dagger \in \mathbb{Z}[\sqrt{2}]$ is totally non-negative

Table 2: Exchange rate α (committed T gates per bit of memory forgone) in several synthesis models.

Model		Rigorous lower bound	Achievable	Granularity
CW Clifford+ T	ancilla-free	1 (Thm. 1); 2 (Thm. 3, given (R)); $1 + \beta \rightarrow 3$ under Conj. 1 (Thm. 5)	$\tau_{RS}/\log_2 K \rightarrow 3$ (Thm. 4); ex- act 2.48 ($L = 5$), 2.72 ($L = 6$); 3.113 ± 0.001 by gridsynth at $\varepsilon = 10^{-10}$	one rotation (fee $2L -$ $O(1)$) under Conj. 1
CW RUS/fallback/mixing (one ancilla, measure- ment)	+	not covered: the count concerns de- terministic words, not channels	$\tau/\log_2 K$, $\tau \approx 1.15L$ [16], smaller with mixed fallback [17]	one rotation (heuristic)
Phase gradient + adder, $Q = 2^b$		none for general prefixes	≈ 4 [18]: commit the high $b - \ell$ bits of a share into the top gradi- ent bits, remember the low ℓ bits, one b -bit addition in the suffix	one bit, no fee: the catalytic gradient state pays the fee once
General Clifford+ T	n -qubit	$(m \log_2 K - S - \log_2 \mathcal{C}_n -$ $O(n))/(2n+1)$ from $\# \leq \mathcal{C}_n (2 \cdot 4^n)^t$; degenerate for $n \sim m$	—	open
$Q \mid 8$		no tradeoff: the family is exactly Clifford or Clifford+ T	—	excluded by $\varepsilon < \sin(\pi/2Q)$

with $N_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(\xi) = \xi\xi^\bullet \leq 2\varepsilon^2 4^k < 1$, which forces $\xi = 0$ and $t = 0$. Then $uu^\dagger = 2^k$; the prime above 2 in $\mathbb{Z}[\omega]$ is $(1 - \omega)$ with $(1 - \omega)^4 \sim 2$, and the units of relative norm one are the ω^j , so $u = \omega^j(1 - \omega)^{2k}$ times a unit and the unitary is $\text{diag}(\omega^a, \omega^b)$ modulo phase. $\square$

B Proof of Theorem 2

Hopf reduction. If $d_{\text{proj}}(W, G_1 R_z(\theta) G_2) \leq \varepsilon$ then, by the closed form of Lemma 1 applied to $W' := G_1^\dagger W G_2^\dagger$, $|W'_{00}| + |W'_{11}| \geq 2 - \varepsilon^2$, hence the off-diagonal entry satisfies $|b(W')|^2 \leq 1 - (1 - \varepsilon^2)^2 \leq 2\varepsilon^2$. Let $H_1 := G_1 S^1 G_1^{-1}$ be the maximal torus containing $G_1 R_z G_1^{-1}$ and $\pi : \text{SU}(2) \rightarrow H_1 \backslash \text{SU}(2) \cong S^2$ the Hopf map. The coset $C = G_1 R_z(\cdot) G_2$ is the fibre through $G_1 G_2$, and $\{W : |b(G_1^\dagger W G_2^\dagger)|^2 \leq 2\varepsilon^2\} = \pi^{-1}(\text{cap})$ with $\mu(\text{cap}) = 2\varepsilon^2$, since under Haar measure $|b|^2$ is uniform on $[0, 1]$. Choose a smooth left- H_1 -invariant majorant $F_+ \geq 1_{\pi^{-1}(\text{cap})}$ supported in $\{|b|^2 \leq 4\varepsilon^2\}$ with $\|\nabla^s F_+\|_\infty \lesssim \varepsilon^{-s}$; then $\mu(F_+) \leq 4\varepsilon^2$ and $\|F_+\|_2 \leq 2\varepsilon$.

Hecke reduction. Λ_k is closed under inversion (T -count is symmetric). Write $\Lambda_k = \bigsqcup_i \gamma_i U$, one left coset per vertex at distance $\leq k$, and let $\bar{F}(g) := \frac{1}{24} \sum_{u \in U} F_+(ug)$, which is left- U -invariant. Then

$$\begin{aligned} \sum_{\gamma \in \Lambda_k} F_+(\gamma) &= \sum_{\gamma \in \Lambda_k} F_+(\gamma^{-1}) = \sum_i \sum_{u \in U} F_+(u^{-1} \gamma_i^{-1}) \\ &= 24 \sum_i \bar{F}(\gamma_i^{-1}) = 24 \sum_{j \leq k} (T_j \bar{F})(e). \end{aligned} \quad (15)$$

Isotypic decomposition. By Peter-Weyl, $L^2(\text{SU}(2)) = \bigoplus_n V_n \otimes V_n^*$, and only even n occur on $\text{PU}(2)$; the decomposition is preserved by left and right translations, hence by T_j and by left- U -averaging. Since F_+ is left- H_1 -invariant, its n -th component is a rank-one matrix coeffi-

cient $F_n(g) = \langle v_0, \pi_n(g) w_n \rangle$ with v_0 the unique H_1 -fixed unit vector and $\|F_n\|_2^2 = \|w_n\|^2/(n+1)$. Left translations act on the v side only, so $T_j \bar{F}_n(g) = \langle v', \pi_n(g) w_n \rangle$ is again rank one, and rank-one coefficients satisfy $\|\phi\|_\infty \leq \sqrt{n+1} \|\phi\|_2$. For $n \geq 1$, using (R) and $\|\bar{F}_n\|_2 \leq \|F_n\|_2$,

$$|(T_j \bar{F}_n)(e)| \leq \sqrt{n+1} \|T_j \bar{F}_n\|_2 \leq \sqrt{n+1} (j+1) 2^{j/2} \|F_n\|_2. \quad (16)$$

The $n = 0$ term contributes $24 \sum_{j \leq k} (\#\text{vertices at distance } j) \mu(F_+) = N_k \mu(F_+) \leq 288 \cdot 2^k \varepsilon^2$.

Summation. $\sum_{j \leq k} (j+1) 2^{j/2} \leq (k+1) 2^{k/2} / (1 - 2^{-1/2}) < 3.42 (k+1) 2^{k/2}$. Split the spectral sum at $M = \lceil c/\varepsilon \rceil$: by Cauchy-Schwarz, $\sum_{n \leq M} \sqrt{n+1} \|F_n\|_2 \leq (\sum_{n \leq M} (n+1))^{1/2} \|F_+\|_2 \lesssim M \varepsilon = O(1)$; for $n > M$, integrating by parts with the Laplacian, $\|F_n\|_2 \leq \|\Delta^s F_+\|_2 / (n(n+1))^s \lesssim \varepsilon^{1-2s} n^{-2s}$, so $\sum_{n > M} \sqrt{n} \|F_n\|_2 \lesssim \varepsilon^{1-2s} M^{3/2-2s} = O(1)$ for $s \geq 1$. Collecting, $\#(\Lambda_k \cap T_\varepsilon(C)) \leq \sum_{\gamma \in \Lambda_k} F_+(\gamma) \leq 288 \cdot 2^k \varepsilon^2 + C_2 (k+1) 2^{k/2}$. $\square$

C Proofs of Theorems 3 and 5

Lemma 5 (Cost-weighted entropy). *Let W take values in an alphabet with cost $t : \mathcal{A} \rightarrow \mathbb{N}$ such that $\#\{W : t(W) \leq \tau\} \leq c_0 + c 2^\tau \varepsilon^\beta$ for all τ . Put $\tau_0 := \lfloor \beta L - \log_2 c \rfloor$, $\mathcal{A}_0 := \{t \leq \tau_0\}$ (so $|\mathcal{A}_0| \leq c_0 + 1$) and $q := \Pr[W \notin \mathcal{A}_0]$. Then*

$$\begin{aligned} H(W) &\leq \mathbb{E}[t(W)] - q(\beta L - \log_2 c - 1) \\ &\quad + h_2(q) + \log_2(c_0 + 1) + 2 \log_2(1 + \mathbb{E}t). \end{aligned} \quad (17)$$

Proof. $H(W) \leq h_2(q) + (1-q) \log_2(c_0 + 1) + q H(W \mid W \notin \mathcal{A}_0)$. On the complement code $t - \tau_0$ in Elias- γ and an

index among at most $(c_0 + 1)c2^t\varepsilon^\beta$ words of cost exactly t (for $t > \tau_0$, $c2^t\varepsilon^\beta \geq 1$): $\text{length} \leq t - \beta L + \log_2 c + \log_2(c_0 + 1) + 2\log_2(t - \tau_0 + 1) + 1$. Then $q\mathbb{E}[t | \notin] \leq \mathbb{E}t$ and $q\log_2(1 + a) \leq \log_2(1 + qa)$. $\square$

Proof of Theorem 3. As in Theorem 1, A is uniform on a product translate given Z , so the A_j are conditionally independent given Z . Per-coordinate Fano (the decoded $\hat{X}_j$ is a function of $(\bar{W}_j^{(t)}, \sigma_t, Z)$ and errs with probability $\leq \delta$): $I(A_j; \bar{W}_j, \sigma_t | Z) \geq \ell_\delta$. By the chain rule and conditional independence,

$$\begin{aligned} m\ell_\delta &\leq \sum_j I(A_j; \sigma_t | Z) + \sum_j I(A_j; \bar{W}_j | \sigma_t, Z) \\ &\leq S + \sum_j I(A_j; \bar{W}_j | \sigma_t, Z), \end{aligned} \quad (18)$$

where $\sum_j I(A_j; \sigma_t | Z) \leq I(A; \sigma_t | Z) \leq H(\sigma_t) \leq S$ because $I(A_j; \sigma | Z, A_{<j}) \geq I(A_j; \sigma | Z)$ under conditional independence.

Fix j and (s, z) , write $t := t_j^{(t)}$, and let E_j be the indicator that coordinate j is correct, $\delta_j := \Pr[E_j = 0 | s, z]$. Then

$$\begin{aligned} &I(A_j; \bar{W}_j | s, z) \\ &\leq h_2(\delta_j) + (1 - \delta_j) I(A_j; \bar{W}_j | E_j = 1, s, z) + \delta_j \log_2 K. \end{aligned} \quad (19)$$

On $E_j = 1$ the alphabet constraint of Corollary 1 holds. Let $L' := L - \log_2(4L) - C'$ so that $\log_2 n(\tau) \leq \tau/2 + \log_2(\tau + 1) + C$ for $\tau \leq 4L'$, and $q' := \Pr[t > 4L' | E_j = 1, s, z]$. Then

$$\begin{aligned} &I(A_j; \bar{W}_j | E_j = 1) \leq h_2(q') \\ &+ (1 - q') \left(\frac{1}{2} \mathbb{E}[t | t \leq 4L'] + \log_2(4L' + 1) + C \right) + q' \log_2 K. \end{aligned} \quad (20)$$

Since $\mathbb{E}[t | E_j = 1] \geq (1 - q')\mathbb{E}[t | t \leq 4L'] + 4L'q'$ and $\log_2 K \leq 2L'$ for large L , the right side is at most $\frac{1}{2}\mathbb{E}[t | E_j = 1] + O(\log L)$. Using $(1 - \delta_j)\mathbb{E}[t | E_j = 1] \leq \mathbb{E}[t | s, z]$,

$$\begin{aligned} &I(A_j; \bar{W}_j | s, z) \\ &\leq \frac{1}{2}\mathbb{E}[t_j | s, z] + h_2(\delta_j) + \delta_j \log_2 K + O(\log L). \end{aligned} \quad (21)$$

Average over (s, z) (Jensen for h_2), sum over j , and use $\sum_j \delta_j \leq m\delta$: $m\ell_\delta \leq S + \frac{1}{2}\bar{T}_t + mh_2(\delta) + \delta m \log_2 K + O(m \log L)$. Rearranging with $\ell_\delta = (1 - \delta) \log_2 K - h_2(\delta)$ gives the claim. $\square$

Proof of Theorem 5. Proceed as above with Conjecture 1 in place of Corollary 1 and Lemma 5 in place of the $\tau/2$ profile, including the E_j split. Two bounds on $\sum_j I(A_j; \bar{W}_j | \sigma_t, Z)$ result. First, with $q_j := \Pr[\bar{W}_j \notin \mathcal{A}_0]$, $I(A_j; \bar{W}_j | E_j = 1, s, z) \leq h_2(q_j) + (1 - q_j) \log_2(c_0 +$

$1) + q_j \log_2 K$; summing with (18) and $\sum_j h_2(\bar{q}_j) \leq m$ gives the bound on $\bar{\mathcal{A}}_t = \sum_j \bar{q}_j$. Second, Lemma 5 gives $\sum_j H(\bar{W}_j | \cdot) \leq \bar{T}_t - (\beta L - \log_2 c - 1)\bar{\mathcal{A}}_t + m + m \log_2(c_0 + 1) + 2m \log_2(1 + \bar{T}_t/m)$, up to the same δ corrections. Inserting the first bound into the second yields the theorem. $\square$

D Proof of Theorem 7

Let Ω be the success event; on Ω , u is a function of the message (snapshot and unitary tuple) together with (v, y) , and the message takes at most $2^{S_{\max}} N_m^{\leq}(T_{\max})$ values; Lemma 2 gives the first inequality. For the second, given Ω let $P' = P \cdot 1_\Omega$, a sub-normalized distribution within δ of P ; then $H_{\max}(U | VY)_{P'} \leq \log_2 \max_{v,y} |\text{supp } P'_{U|v,y}|$, so $H_{\max}^\delta \leq H_0^\delta$. $\square$

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